ESP32 Fiber → 4G Failover

User Manual – English Version

1. Overview

This system automatically switches from a primary fiber Internet connection to a 4G/LTE backup router when the Internet becomes unavailable. It is based on an ESP32-C5 with web interface, LCD display, status LEDs, and relay control.

2. Main Features

1. Automatic Fiber → 4G failover
2. Automatic return to Fiber when Internet is restored
3. Fiber router reset relay after repeated failures
4. Web interface with status and event history
5. LCD 16x2 real-time status display
6. RGB status LED
7. Manual control via push buttons or web interface
8. Wi■Fi configuration portal if connection is lost

3. Operating Modes

AUTO mode:

1. Normal state: Fiber active, 4G router OFF
2. After 3 failed Internet checks: fiber router reset
3. 4G router activated automatically
4. When Internet returns: system switches back to Fiber

MANUAL mode:

1. 4G router controlled only by buttons or web interface
2. No automatic return to Fiber

4. Buttons

1. Short press: activate 4G router
2. Long press: deactivate 4G router
3. Short press on any button: wake LCD display

5. LCD Display

1. Line 1: Fiber status and Wi■Fi band (2.4G or 5G)
2. Line 2: Scrolling SSID and ESP32 IP address
3. Backlight turns off after inactivity
4. Backlight turns on after button press or web access

6. Web Interface

1. Real-time Fiber and 4G status
2. Operating mode (AUTO or MANUAL)
3. Connected SSID and Wi■Fi band
4. ESP32 IP address
5. Event history log
6. Control buttons for all functions

7. First Setup

1. Power the system.
2. If no Wi■Fi is stored, the ESP32 starts configuration mode.
3. Connect to Wi■Fi network 'Config-Monitor'.
4. Open browser at 192.168.4.1.
5. Enter Wi■Fi SSID and password.
6. Device reboots and connects to your network.

8. Normal Operation

1. Device connects automatically to Wi■Fi.
2. LCD shows status information.
3. Web interface available at ESP32 IP.
4. System manages failover automatically.

9. Power Outage Behavior

After a power outage, the ESP32 reconnects to the last known Wi■Fi network. If connection fails, it starts the configuration portal.

10. Safety Notes

1. Use a certified AC■DC power module.
2. Ensure insulation for all 230V connections.
3. Install inside a closed, non■conductive enclosure.
4. Intended for experienced users.